

CSE563 — Assignment 1

Unit I: Basic Mathematics in Computer Science

Maps to CO1 (recall the basic concepts of mathematics in computer science and engineering).
Marks: 50. Attempt all questions. Show every step of working; an unsupported final answer earns no marks.

Part A — Short Answer and Definitions (10 marks, 2 each)

Q1. [2 marks] State the division algorithm. Using it, find q and r when $a = -47$ and $d = 9$, and explain why the C expression `-47 % 9` does not return your value of r .

Solution. Division algorithm: for any integer a and positive integer d there exist unique integers q, r with $a = dq + r$ and $0 \leq r < d$.
 $-47 = 9(-6) + 7$, since $9(-6) = -54$ and $-54 + 7 = -47$, with $0 \leq 7 < 9$. So $q = -6, r = 7$.
C (and Java, C++) define `%` so that the remainder takes the sign of the *dividend*, truncating the quotient toward zero: $-47/9 = -5$ (truncated), giving $-47 - 9(-5) = -2$. So `-47 % 9` is -2 , not 7 . Python uses floor division and returns 7 . Normalise in C with `((a % d) + d) % d`.

Q2. [2 marks] Let $A = \{1, 2, 3\}$. Write $\mathcal{P}(A)$ in full and state $|\mathcal{P}(A)|$. If $|B| = 6$, how many proper subsets does B have?

Solution. $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$, so $|\mathcal{P}(A)| = 2^3 = 8$.
 B has $2^6 = 64$ subsets, of which all but B itself are proper: 63 proper subsets.

Q3. [2 marks] Write the contrapositive, converse and inverse of: “If a graph is a tree, then it is connected.” State which of the three is logically equivalent to the original.

Solution. Let $p =$ “the graph is a tree”, $q =$ “it is connected”.

- **Contrapositive** ($\neg q \rightarrow \neg p$): if a graph is not connected, then it is not a tree. **Equivalent to the original.**
- **Converse** ($q \rightarrow p$): if a graph is connected, then it is a tree. (False — a cycle is connected but not a tree.)
- **Inverse** ($\neg p \rightarrow \neg q$): if a graph is not a tree, then it is not connected. (Also false; equivalent to the converse.)

Q4. [2 marks] Define an equivalence relation. Show that congruence modulo 4 on $\mathbb{Z}$ is one, and list its equivalence classes.

Solution. An equivalence relation is reflexive, symmetric and transitive.

- **Reflexive:** $4 \mid (a - a) = 0$, so $a \equiv a$.
- **Symmetric:** if $4 \mid (a - b)$ then $4 \mid -(a - b) = (b - a)$.
- **Transitive:** if $a - b = 4k$ and $b - c = 4m$ then $a - c = 4(k + m)$.

Classes: $[0] = \{\dots, -4, 0, 4, 8, \dots\}$, $[1], [2], [3]$ — four classes partitioning $\mathbb{Z}$.

Q5. [2 marks] Classify $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 - 2$ as injective / surjective / bijective, with justification. Give f^{-1} if it exists.

Solution. Injective: $a^3 - 2 = b^3 - 2 \Rightarrow a^3 = b^3 \Rightarrow a = b$ (the cube function is strictly increasing on $\mathbb{R}$). ✓

Surjective: for any $y \in \mathbb{R}$, $x = \sqrt[3]{y+2}$ is a real number with $f(x) = y$. ✓
Therefore **bijective**, with $f^{-1}(y) = \sqrt[3]{y+2}$.

Part B — Numerical Problems (24 marks)

Q6. [5 marks] Use the Euclidean algorithm to compute $\gcd(1071, 462)$, then use the extended Euclidean algorithm to find integers x, y with $1071x + 462y = \gcd(1071, 462)$. Verify your answer. Finally state $\text{lcm}(1071, 462)$.

Solution.

$$1071 = 2(462) + 147$$

$$462 = 3(147) + 21$$

$$147 = 7(21) + 0$$

So $\gcd = 21$.

Back-substitution:

$$\begin{aligned} 21 &= 462 - 3(147) \\ &= 462 - 3(1071 - 2(462)) \\ &= 462 - 3(1071) + 6(462) = 7(462) - 3(1071). \end{aligned}$$

So $x = -3$, $y = 7$. Verify: $-3(1071) + 7(462) = -3213 + 3234 = 21$. ✓

$$\text{lcm} = \frac{1071 \times 462}{21} = \frac{494802}{21} = \mathbf{23562}.$$

Q7. [5 marks] (a) Find $11^{-1} \pmod{26}$ using the extended Euclidean algorithm. (b) Compute $5^{20} \pmod{7}$ using Fermat's little theorem, and verify by fast modular exponentiation.

Solution. (a) $\gcd(26, 11)$:

$$26 = 2(11) + 4$$

$$11 = 2(4) + 3$$

$$4 = 1(3) + 1$$

Back-substitute: $1 = 4 - 3 = 4 - (11 - 2 \cdot 4) = 3(4) - 11 = 3(26 - 2 \cdot 11) - 11 = 3(26) - 7(11)$. So $-7 \cdot 11 \equiv 1 \pmod{26}$, and $-7 \equiv 19$. Check: $11 \times 19 = 209 = 8(26) + 1$. ✓ $11^{-1} \equiv \mathbf{19} \pmod{26}$.

(b) 7 is prime and $7 \nmid 5$, so by Fermat $5^6 \equiv 1 \pmod{7}$. Since $20 = 6(3) + 2$:

$$5^{20} = (5^6)^3 \cdot 5^2 \equiv 1^3 \cdot 25 \equiv 25 - 21 = \mathbf{4} \pmod{7}.$$

Verify by square-and-multiply: $20 = 10100_2$. $5^1 \equiv 5$; $5^2 \equiv 25 \equiv 4$; $5^4 \equiv 4^2 = 16 \equiv 2$; $5^8 \equiv 2^2 = 4$; $5^{16} \equiv 4^2 = 16 \equiv 2$. Then $5^{20} = 5^{16} \cdot 5^4 \equiv 2 \times 2 = 4$. ✓

Q8. [4 marks] In a survey of 120 students: 65 use Windows, 48 use Linux, 40 use macOS, 22 use Windows and Linux, 18 use Windows and macOS, 15 use Linux and macOS, and 8 use all three. How many use (a) at least one of the three, (b) none of them, (c) exactly one?

Solution. Inclusion–exclusion:

$$|W \cup L \cup M| = 65 + 48 + 40 - 22 - 18 - 15 + 8 = \mathbf{106}.$$

(b) None: $120 - 106 = \mathbf{14}$.

(c) Exactly one = $|W \cup L \cup M| - (\text{exactly two}) - (\text{exactly three})$. Exactly three = 8. Exactly two = $(22 - 8) + (18 - 8) + (15 - 8) = 14 + 10 + 7 = 31$. Exactly one = $106 - 31 - 8 = \mathbf{67}$. (Check directly: Windows only = $65 - 14 - 10 - 8 = 33$; Linux only = $48 - 14 - 7 - 8 = 19$; macOS only = $40 - 10 - 7 - 8 = 15$; $33 + 19 + 15 = 67$. ✓)

Q9. [5 marks] Simplify $F(A, B, C) = \sum m(0, 2, 4, 5, 6)$ using a Karnaugh map. Draw the map, mark your groups, and write the minimal SOP expression. Verify your answer algebraically for the input $A=1, B=0, C=1$.

Solution.

	BC=00	01	11	10
A=0	1	0	0	1
A=1	1	1	0	1

red (wraps): $\overline{C}$
gold: $A\overline{B}$

Minterms present: $m_0 = \overline{A}\overline{B}\overline{C}$, $m_2 = \overline{A}B\overline{C}$, $m_4 = A\overline{B}\overline{C}$, $m_5 = A\overline{B}C$, $m_6 = ABC$.

Groups:

- **Red** — a group of four that wraps around the left and right edges: columns $BC=00$ and $BC=10$, both rows, i.e. cells m_0, m_4, m_2, m_6 . In all four, $C = 0$ while A and B each take both values, so both variables cancel $\Rightarrow \overline{C}$.
- **Gold** — a group of two: row $A = 1$, columns 00 and 01, i.e. m_4, m_5 . Here $A = 1$, $B = 0$, and C varies $\Rightarrow A\overline{B}$.

Every 1 is covered (m_5 only by the gold group, so both groups are essential). **Minimal SOP:** $F = \overline{C} + A\overline{B}$.

Verification at $A=1, B=0, C=1$ (minterm m_5 , which should be 1): $\overline{C} = 0$; $A\overline{B} = 1 \cdot 1 = 1$; $F = 0 + 1 = 1$. ✓ And at $A=0, B=1, C=1$ (m_3 , should be 0): $\overline{C} = 0$, $A\overline{B} = 0$, $F = 0$. ✓

Q10. [5 marks] For $f(x, y) = x^2 + 3xy + 2y^2$: (a) find $\partial f / \partial x$ and $\partial f / \partial y$; (b) evaluate the gradient at $(1, 2)$; (c) perform one step of gradient descent from $(1, 2)$ with learning rate $\alpha = 0.1$ and state whether f decreased.

Solution. (a) $\frac{\partial f}{\partial x} = 2x + 3y$; $\frac{\partial f}{\partial y} = 3x + 4y$.

(b) At $(1, 2)$: $\nabla f = (2 + 6, 3 + 8) = (\mathbf{8, 11})$.

(c) $(x, y) \leftarrow (1, 2) - 0.1(8, 11) = (1 - 0.8, 2 - 1.1) = (\mathbf{0.2, 0.9})$.

$f(1, 2) = 1 + 6 + 8 = 15$. $f(0.2, 0.9) = 0.04 + 3(0.18) + 2(0.81) = 0.04 + 0.54 + 1.62 = 2.20$.

The value fell from 15 to 2.20, so this step decreased f . ✓

Part C — Proofs and Applied Reasoning (16 marks)

Q11. [5 marks] Prove by mathematical induction that $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$. State the basis step and the inductive step explicitly.

Solution. Basis ($n = 1$): LHS = $1^2 = 1$; RHS = $\frac{1 \cdot 2 \cdot 3}{6} = 1$. ✓

Inductive hypothesis: assume for some $k \geq 1$ that $\sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$.

Inductive step:

$$\begin{aligned} \sum_{i=1}^{k+1} i^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\ &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} = \frac{(k+1)(2k^2 + 7k + 6)}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6} = \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}, \end{aligned}$$

which is the formula at $n = k + 1$. By the principle of mathematical induction the identity holds for all $n \geq 1$. ■

Q12. [4 marks] Prove by induction that $3 \mid (n^3 - n)$ for every integer $n \geq 0$.

Solution. Basis ($n = 0$): $0^3 - 0 = 0 = 3(0)$. ✓

Inductive hypothesis: assume $k^3 - k = 3m$ for some integer m .

Inductive step:

$$(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k + 1 - k - 1 = (k^3 - k) + 3k^2 + 3k = 3m + 3(k^2 + k) = 3(m + k^2 + k),$$

which is divisible by 3. Hence $3 \mid (n^3 - n)$ for all $n \geq 0$. ■

(Alternative one-line argument: $n^3 - n = (n-1)n(n+1)$, a product of three consecutive integers, one of which must be a multiple of 3.)

Q13. [4 marks] Prove the set identity $A - (B \cap C) = (A - B) \cup (A - C)$ by the element method (double inclusion). Name the logical law that does the work.

Solution. ($\subseteq$) Let $x \in A - (B \cap C)$. Then $x \in A$ and $x \notin (B \cap C)$, i.e. $\neg(x \in B \wedge x \in C)$. By **De Morgan's law**, $x \notin B \vee x \notin C$.

- If $x \notin B$: since $x \in A$, $x \in A - B$.
- If $x \notin C$: since $x \in A$, $x \in A - C$.

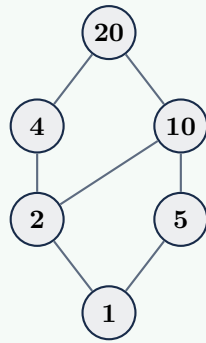
Either way $x \in (A - B) \cup (A - C)$.

($\supseteq$) Let $x \in (A - B) \cup (A - C)$. Then $x \in A$ in both cases, and either $x \notin B$ or $x \notin C$, so $\neg(x \in B \wedge x \in C)$, i.e. $x \notin B \cap C$. Hence $x \in A - (B \cap C)$.

Both inclusions hold, so the sets are equal. The law used is **De Morgan's**: $\neg(p \wedge q) \equiv \neg p \vee \neg q$. ■

Q14. [3 marks] Draw the Hasse diagram of the poset $(D_{20}, |)$ where $D_{20} = \{1, 2, 4, 5, 10, 20\}$ is the set of positive divisors of 20. Identify the least and greatest elements, and give one pair of incomparable elements together with their greatest lower bound and least upper bound.

Solution.



Least element: 1 (divides everything). **Greatest element:** 20 (divisible by everything).

Incomparable pair: 4 and 5 (neither divides the other). Also 4 and 10.

- $\text{glb}(4, 5) = \text{gcd}(4, 5) = 1$; $\text{lub}(4, 5) = \text{lcm}(4, 5) = 20$.

Since every pair has a glb and a lub, this poset is a lattice. Note there is no edge $2 \rightarrow 20$: it is implied by transitivity through 4 or 10, and Hasse diagrams omit implied edges.